

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Eight Semester of B. Tech. Examination (EE)

May 2014

EE-408 Power System Stability and Control

Date: 09/05/2014 Friday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Define Stability and give the Classification of Power system stability in detail. [07]

Q - 2 (a) Derive the swing equation and discuss M and H Constant related to stability. [07]

(b) A synchronous generator connected to a 50 Hz infinite bus-bar via a transformer and transmission line has an inertia constant of 3.5 MJ/MVA. The synchronous reactance of the generator is 1.2 p.u. and total reactance of the transformer and transmission line is 0.8 p.u. If the no load voltage of the generator is 1.15 p.u.

Calculate:

(1) The frequency of natural oscillation of the generator rotor if the generator is loaded to 50% of its maximum power limit.

(2) The frequency of natural oscillation of the generator rotor if the generator is loaded to 75% of its maximum power limit.

OR

Q - 2 (a) Explain the modeling of synchronous generator. Also discuss the saliency and non-saliency effect. [07]

(b) Define the following term: [04]

- (1) Eigen vector (2) Voltage security (3) Voltage collapse proximity indicator
(4) Synchronizing coefficient

(c) Explain the operating states of power system in detail. [03]

Q - 3 (a) Explain the effect of sudden loss of one of parallel line on Equal area criterion. [06]

(b) What do you mean by Transient stability? Derive the condition of stability for the system. [05]

(c) Give the factors affecting Transient stability. [03]

OR

- Q - 3 (a) Derive the equation of critical clearing time and critical clearing angle for transient stability with necessary diagram. [07]
- (b) Find the critical clearing angle for the system as shown in Fig.1. For a three-phase fault at the point P. The generator is delivering 1.0 p.u. power under prefault conditions. [07]

SECTION – II

- Q - 4 (a) What do you mean by Voltage stability? Derive the equation of voltage stability limit. [07]
- Q - 5 (a) Explain the PV and QV curve related to voltage stability with necessary diagram. [10]
- (b) Explain the characteristic of Reactive power compensating devices for voltage stability. [04]

OR

- Q - 5 (a) Explain the block diagram representation of single machine infinite bus system with classical generator model. [07]
- (b) Explain the power system stabilizer with necessary block diagram. [07]
- Q-6 (a) Explain the different methods for enhancement of transient stability. [07]
- (b) Explain the operating principle of TCSC with necessary diagram. [07]

OR

- Q-6 (a) Explain the operating principle of STATCOM along with its V-I characteristic. [07]
- (b) How the steady state power transfer capability can be increase with SVC? [04]
- (c) Explain the corrective steps for prevention of voltage collapse. [03]

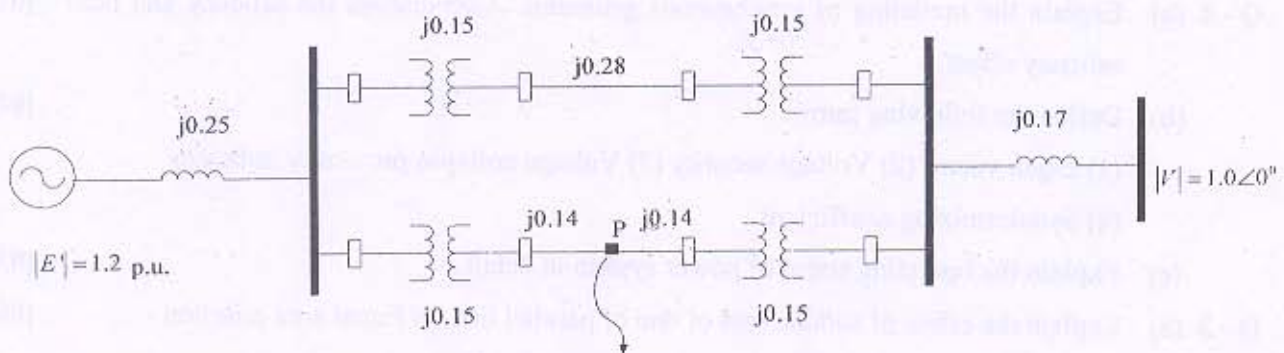


Fig.1

BEST OF LUCK
